EXPERIMENT 23
Text Coherence Evaluation

Aim

To evaluate simple text coherence using sentence-to-sentence word overlap.

Procedure

1. Prepare a short multi-sentence text.
2. Split the text into sentences.
3. Convert each sentence to a set of words.
4. Calculate overlap between adjacent sentence word sets.
5. Display average overlap as a simple coherence indicator.

Program

```
import re

text = ("Python is a programming language. "
        "Python supports many programming tasks. "
        "Many tasks require careful planning.")
sentences = [set(re.findall(r"\b\w+\b", s.lower()))
             for s in re.split(r"(?<=[.!?])\s+", text)]
scores = []
for first, second in zip(sentences, sentences[1:]):
    union = first | second
    scores.append(len(first & second) / len(union) if union else 0)

average = sum(scores) / len(scores) if scores else 0
print("Adjacent sentence overlaps:", [round(s, 3) for s in scores])
print("Simple coherence score:", round(average, 3))
```

Output

```
OUTPUT — EXPERIMENT 23
Adjacent sentence overlaps: [a numeric score, a numeric score]
Simple coherence score: a numeric score
(The program calculates the exact scores when run.)
```

Result

Thus, adjacent-sentence word overlap was calculated as a basic indicator of text coherence.